When Auxiliary Losses Fail: Non-Stationary Targets Induce Directional Gradient Noise

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Auxiliary losses often stabilize representation learning yet sometimes degrade late-training performance in ways that look like ordinary RL noise. We identify the learning-dynamics mechanism: when auxiliary targets are both *structured* (coupled to the task state) and *non-stationary* (drifting as other learners update), they inject directional gradient noise whose contribution to parameter variance dominates that of the vanishing policy gradient near convergence. The effect is not magnitude imbalance, not multi-task conflict, and not capacity consumption; it is driven by target drift across learners. A target-source distinguishing test in cooperative MARL holds aux capacity, architecture, and the aux-to-encoder gradient pathway fixed and varies only the target: co-learning teammate targets are pathological ($d = +1.40$ vs No Aux, bootstrap CI above zero), while structured-but-stationary (frozen-initial-policy) and unstructured-stationary (random) targets are both not resolved from No Aux at n=5 (CIs cross zero). A complementary $8\times$ aux-capacity sweep has no effect, jointly inconsistent with capacity-consumption explanations. Effect size scales with coordination-induced drift across four MARL benchmarks; supervised CIFAR-100 marks the stationarity boundary at near-zero effect, as the theory predicts. Three gradient-pathway interventions (λ-annealing, stop-gradient on the belief, critic-side aux placement) recover the no-aux regime on Overcooked.

1 Introduction

Auxiliary prediction losses are ubiquitous in modern deep learning: UNREAL [24], world models [20], self-predictive representations [45], and the MARL belief-learning pattern of Dynamic Belief [60] and BEPAL [22] all attach a constant-weight auxiliary loss to a shared encoder. Most of the time it helps. Sometimes it silently corrupts late-training convergence with cross-seed variance that looks like ordinary RL noise. This paper identifies the learning-dynamics principle that separates the two regimes.

Principle (informal). *Auxiliary learning fails most severely when targets are both structured and co-adaptive, because the resulting gradient direction becomes a non-stationary process that overwhelms the vanishing policy gradient near convergence; removing either ingredient reduces the effect.*

What it is, what it is not. The failure mechanism is *directional*, but distinct from the persistent-conflict patterns existing gradient-surgery methods correct: GradNorm rebalances magnitudes [8] and PCGrad projects out persistent inter-task conflict directions [59]. Neither addresses temporal variance in alignment between two gradients whose mean is zero, which is the signature we observe (Figure 2b). Note also that the aux gradient's norm stays strictly below the policy gradient's (Figure 2c). It is not multi-task conflict, because a single auxiliary is enough. And it is not capacity consumption, because an $8\times$ sweep of aux capacity has no effect. What matters is the *target*: when it drifts with another

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

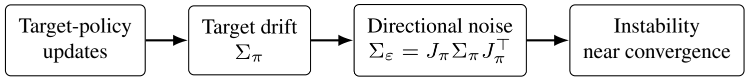


Figure 1: Causal chain from a co-adaptive target-generating policy to late-training instability. The Jacobian J_π is a fixed sensitivity of the aux gradient to the target policy (Step 2); the drift covariance Σ_π is the time-varying input. Eq. (5) combines them; Proposition 1 closes the chain to parameter variance.

Table 1: The principle we identify versus adjacent prior work.

	Existing	**This paper**
Mechanism	magnitude interference	**directional interference**
Targets	static / stationary	**non-stationary**
Setting	multi-task conflict	**single-task + auxiliary**
Driver	gradient norm imbalance	**target drift across learners**

learner's updates, the aux-gradient direction becomes a non-stationary process that shrinking policy gradients near convergence cannot dominate.

Cooperative MARL as test bed. Co-learning teammates are a clean natural source of structured non-stationary targets: a teammate-action-prediction aux target is coupled to the game state *and* drifts as teammates update. We construct **VABL** (§3) as a minimal architecture in this class and test the principle with three experiments. *(i) Target-source distinguishing test* (§6.2): the principle predicts the ordering random $<$ frozen $<$ teammate in pathology severity; the data confirm it. *(ii) Aux-capacity sweep*: $8\times$ variation has no effect. *(iii) Boundary*: pathology scales with coordination-induced teammate drift across four MARL benchmarks; MPE's slow symmetric drift and supervised CIFAR-100 show zero effect as the principle predicts for stationary targets.

Contributions.

- **Principle.** A learning-dynamics failure mode: structured, non-stationary auxiliary targets induce directional gradient noise near convergence, with a testable ordering prediction over target sources (random $<$ frozen $<$ co-learning).
- **Mechanism and theory.** A target-source intervention and an $8\times$ aux-capacity sweep are jointly inconsistent with capacity-consumption and multi-task-conflict explanations; a first-order model links target-policy drift to gradient noise via $\Sigma_\varepsilon \approx J_\pi \Sigma_\pi J_\pi^\top$ (Eq. (5)), with a drift-to-variance bound and a cosine-std diagnostic that separates pathway-active from pathway-zero conditions.
- **Validation.** The predicted ordering and null conditions hold across four cooperative MARL benchmarks, a supervised-stationary boundary (CIFAR-100), and a slow-symmetric-drift boundary (MPE simple_spread); reported on a 290+-run matrix with 5 seeds and bootstrap CIs.
- **Practical fixes.** Three gradient-pathway interventions (λ-annealing, stop-gradient on the belief, critic-side aux placement) recover the no-aux regime on Overcooked; SMAX retains a residual mean gap with non-monotonic λ response that we flag as open.

2 Related Work

Auxiliary losses in RL. UNREAL [24] pioneered auxiliary tasks; subsequent work uses world-model losses [20], self-predictive representations [45, 16], and contrastive objectives [27]. Theoretical analyses of auxiliary-task dynamics [32, 51] cover TD and latent self-prediction; [14, 29] argue constant weights are suboptimal. [34] document on-policy PPO representation collapse under non-stationarity; [10] show successor-feature losses collapse basis features to cosine-to-1, resolved by stop-gradient in the reward loss (an analogue to our stop-gradient fix path). In MARL, XP-MARL [56] prioritizes training samples for non-stationarity; BEPAL [22] and Dynamic Belief [60] directly instantiate the design pattern we study (attention-based belief encoder + fixed-weight teammate-future-prediction aux). Zhai et al. [60]'s Static-Belief ablation (their Table 2) reports that naive co-training drops Traffic Junction success from 73.4% to 71.9%; their fix is a meta-learned dynamic context, while we identify the mechanism and show pathway-severing interventions suffice without meta-learning. BEPAL's oracle (stationary) targets are consistent with the principle. Contemporaneous [53] identifies a distinct magnitude-driven pathology in parameter-shared DQN.

Attention and stop-gradient. AERIAL [39] uses QMIX-based attention over teammate hidden states (no auxiliary head); MAAC [23] places attention in the centralized critic; MAT [52] uses an encoder-decoder for centralized policy learning. None pair attention-based *decentralized* belief encoding with a constant-weight teammate-future-prediction aux loss. In SSL, BYOL [19], SimSiam [7], and DINO [5] use stop-gradient to prevent representation collapse; our stop-gradient fix path is the MARL analogue.

Gradient interference and coordination collapse. PCGrad [59] and GradNorm [8] project or reweight gradients by inter-task magnitude or persistent conflict direction; multi-objective and multi-agent variants [54, 61, 48, 35] extend this to direction-aware scheduling and cross-agent decoupling. Closest to our diagnostic, DG-PG [57] decomposes cooperative-MARL gradient variance into self vs. cross-agent components with a closed-form combiner; our drift-gated approximation (§7) targets the same object on the temporal policy-vs-auxiliary axis within a single agent. Our pathology is intra-agent, directional rather than magnitude-driven (Figure 2c), and environment-dependent. Overcooked [6] is widely studied with prior reports of severe degradation [58, 40] and budget sensitivity [17]; the residual instability we identify at 10M is distinct and gradient-mediated.

3 Background: Attention-Based Belief Learning

The design pattern we investigate, attention-based belief encoders trained with constant-weight auxiliary prediction losses targeting teammate future behavior, has recent instances in BEPAL [22] and Dynamic Belief [60], and sits within the broader family of constant-weight aux architectures [20, 45, 24]. We construct **VABL** as a minimal cooperative Dec-POMDP [36] instance of this class in which attention, aux loss, and aux-to-encoder gradient flow can each be manipulated independently.

Belief update. Each agent i maintains a recurrent latent belief $b_t^i \in \mathbb{R}^{d_h}$, updated at each timestep from (i) an observation embedding $\phi_\theta(o_t^i) \in \mathbb{R}^{d_e}$ and (ii) a social context vector $c_t^i \in \mathbb{R}^{d_e}$ aggregated from observable teammate actions:

$$b_t^i = \mathrm{GRU}\left([\phi_\theta(o_t^i) \| c_t^i],\ b_{t-1}^i\right), \tag{1}$$

where GRU denotes a gated recurrent unit [9].

Attention context. The social context c_t^i is computed via multi-head attention [50] over action embeddings of visible teammates, with query $W_q b_{t-1}^i$ and keys/values $\{\psi_\theta(a_{t-1}^j) + e_j\}_{j \neq i}$ (learnable identity embeddings e_j); per-step visibility masks zero out invisible teammates. The mean-pooling ablation replaces MHA with the unweighted average.

Policy and auxiliary heads. The policy $\pi_\theta(a_t^i \mid b_t^i)$ is a linear head on the belief. The auxiliary predictor $\hat{\pi}_\varphi(\cdot \mid b_t^i)$ is a 2-layer MLP that predicts each visible teammate's next action from the current belief, trained with cross-entropy:

$$\mathcal{L}_{\mathrm{aux}}^i = \mathbb{E}\left[\sum_{j \neq i} m_{t+1}^{i \leftarrow j}\, \mathrm{CE}(\hat{\pi}_\varphi(\cdot \mid b_t^i),\ a_{t+1}^j)\right]. \tag{2}$$

Training objective. The total loss combines a PPO policy gradient [44], a centralized value loss, entropy regularization, and the auxiliary loss:

$$\mathcal{L} = \mathcal{L}_{\mathrm{PPO}} + c_v \mathcal{L}_{\mathrm{value}} + c_e \mathcal{L}_{\mathrm{entropy}} + \lambda \sum_i \mathcal{L}_{\mathrm{aux}}^i, \tag{3}$$

where $\lambda = 0.05$ is the auxiliary loss weight. In the standard configuration, λ is held constant throughout training.

Gradient pathway. The auxiliary loss gradient flows through the belief b_t^i into the attention mechanism, GRU, and observation encoder ϕ_θ. Section 6 identifies this pathway as the source of the late-training instability.

4 Theory: From Target-Policy Drift to Directional Interference

Intuitively, as policy gradients shrink near convergence, even small fluctuations in auxiliary gradient direction can dominate updates, turning convergence into a stochastic drift process. We derive a first-order mechanistic model of this effect in four steps: (i) define directional interference and distinguish it from magnitude interference; (ii) derive the aux-gradient noise covariance from target-policy drift; (iii) bound late-training parameter variance in terms of that covariance; (iv) state a threshold condition for instability onset. The model connects co-learning dynamics to observed instability through causally upstream quantities (Σ_π, J_π) rather than treating gradient noise as exogenous, using standard linearization steps we make explicit below.

Step 1: Defining directional interference. Let $g_\pi(t) = \nabla_\theta L_{\text{PPO}}(\theta_t)$ and $g_{\text{aux}}(t) = \lambda \nabla_\theta L_{\text{aux}}(\theta_t; \pi_j^{(t)})$ be the two gradient signals the actor receives at iteration t. Prior gradient-conflict work (PCGrad [59], GradNorm [8]) treats conflict as a *magnitude* imbalance: persistent $\mathbb{E}[g_\pi^\top g_{\text{aux}}] < 0$ or $\|g_{\text{aux}}\| \gg \|g_\pi\|$. We define *directional interference* as temporal variance in gradient alignment:

$$I_{\text{dir}} = \text{Var}_t\left[\frac{g_\pi(t)^\top g_{\text{aux}}(t)}{\|g_\pi(t)\|\|g_{\text{aux}}(t)\|}\right], \quad I_{\text{mag}} = \left(\mathbb{E}_t\left[\frac{g_\pi(t)^\top g_{\text{aux}}(t)}{\|g_\pi(t)\|\|g_{\text{aux}}(t)\|}\right]\right)^2. \tag{4}$$

High I_{mag} signals a persistent non-zero alignment between the two gradients (the sign of $\mathbb{E}_t[\cos]$ tells us whether persistent agreement or persistent conflict; PCGrad projects out the conflict case). High I_{dir} with $I_{\text{mag}} \approx 0$ signals alignment that *oscillates around zero*: neither helpful nor persistently harmful but corrosive to convergence. Figure 2 measures Full VABL's late-training signature ($\mathbb{E}[\cos] \approx 0, \text{std}(\cos) = 0.185$), i.e. $I_{\text{dir}} \gg I_{\text{mag}}$.

Step 2: Deriving Σ_ε from target-policy drift. Let $\pi_j \in \mathbb{R}^{d_\pi}$ denote the policy whose actions define the aux label at step t; in Full VABL this is the teammate's own co-learning policy, while the frozen-target and random-target conditions (§5.2) replace it with a stationary alternative, so that Σ_π and J_π below always refer to the target-generating policy rather than to any other agent. Represent π_j either by its parameters or, equivalently in the linear regime, by its action-probability vector stacked over the observed state distribution. Decompose the aux gradient as $g_{\text{aux}}(t) = \bar{g}_{\text{aux}} + \varepsilon(t)$. A first-order expansion of g_{aux} around $\bar{\pi}_j$, the time-averaged target-generating policy over the analysis window, gives

$$\varepsilon(t) \approx J_\pi\,(\pi_j^{(t)} - \bar{\pi}_j), \qquad \Sigma_\varepsilon \approx J_\pi\,\Sigma_\pi\,J_\pi^\top, \qquad \Sigma_\pi = \text{Cov}_t[\pi_j^{(t)}], \tag{5}$$

where $J_\pi \in \mathbb{R}^{|\theta| \times d_\pi} = \partial g_{\text{aux}}/\partial \pi_j|_{\bar{\pi}_j}$ and J_π absorbs the λ scaling of g_{aux}. The gradient noise covariance factors through the target-generating policy's drift covariance. Two limits: a frozen target-generating policy ($\Sigma_\pi = 0$) gives $\Sigma_\varepsilon \to 0$; a target constructed not to depend on the target-generating policy ($J_\pi = 0$, e.g. random or supervised targets) also gives $\Sigma_\varepsilon \to 0$. The target-source distinguishing test (§6.2) realizes both limits empirically. Equivalently, DG-PG's variance-reduction framework for cooperative MARL [57] assumes an *exogenous* guidance target, $\nabla_\theta \tilde{x} = 0$ (their Assumption 3.1); in our setting the aux target is generated by a co-evolving policy (the teammate's in Full VABL), so this exogeneity assumption is violated and Σ_ε is driven by Σ_π rather than collapsing. The δ-stationarity bound of [28] formalizes Σ_π via the KL-divergence of consecutive joint policies. The linearization is local in policy space and tightest in late training, where convergence implies small per-iteration drift.

Step 3: Noise-driven parameter variance. Near a local optimum θ^* the policy loss is approximately quadratic with Hessian $H \succ 0$, giving $\nabla_\theta L_{\text{PPO}} = H(\theta - \theta^*)$ which shrinks as $\theta \to \theta^*$. Centering around $\theta^*_{\text{eff}} = \theta^* - H^{-1}\bar{g}_{\text{aux}}$, the displacement $\delta(t) = \theta(t) - \theta^*_{\text{eff}}$ evolves as

$$\delta(t{+}1) = (I - \alpha H)\,\delta(t) - \alpha\,\varepsilon(t). \tag{6}$$

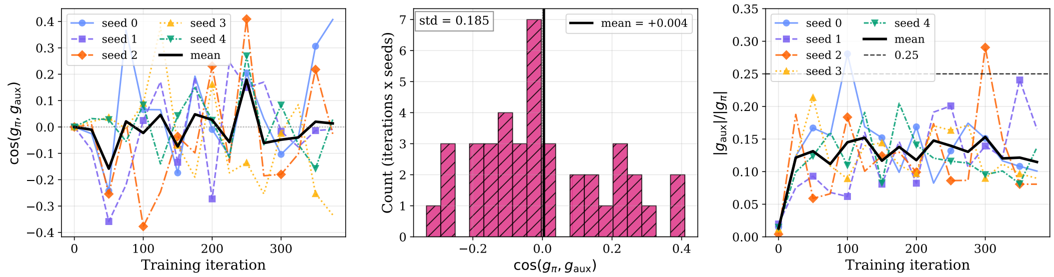


Figure 2: Gradient decomposition for Full VABL on Overcooked AA (5 seeds). **(a)** Per-seed cosine similarity between the PPO and auxiliary gradients oscillates across both signs throughout training; across-seed mean in bold black. **(b)** Cosine values pooled across seeds over the late 50% of training: mean ≈ 0, std $= 0.185$, range $[-0.33, +0.41]$. Symmetric spread consistent with directional noise in the aux-gradient drift. **(c)** The aux-to-policy magnitude ratio mean stays bounded below 0.25 across training (one seed touches 0.29 briefly mid-training), supporting the directional-not-magnitude interpretation.

Proposition 1 (Local drift-to-variance scaling)**.** *Under Eq.* (6) *with $\varepsilon(t)$ treated as i.i.d. with covariance given by Eq.* (5) *and spectral radius $\rho(I - \alpha H) < 1$, the steady-state displacement variance satisfies*

$$\mathbb{E}[\|\delta\|^2] \;\propto\; \frac{\alpha\,\mathrm{tr}(J_\pi\,\Sigma_\pi\,J_\pi^\top)}{\lambda_{\min}(H)}. \tag{7}$$

The bound gives the scaling *shape*, not absolute magnitude: colored $\varepsilon(t)$ inflates the right-hand side by an autocorrelation-time factor τ_c without changing the dependence on $\mathrm{tr}(J_\pi \Sigma_\pi J_\pi^\top)$ or $\lambda_{\min}(H)$.

Step 4: threshold and testable predictions. Let $\eta = \mathbb{E}_t[\|g_\pi(t)\|]$ denote the typical policy-gradient magnitude; the same Lyapunov argument behind Proposition 1 gives a policy-signal displacement variance $\alpha\eta^2/\lambda_{\min}(H)$. Instability onset corresponds to the aux-drift variance exceeding this signal variance:

$$\underbrace{\frac{\alpha\,\mathrm{tr}(J_\pi\,\Sigma_\pi\,J_\pi^\top)}{\lambda_{\min}(H)}}_{\text{aux-drift variance}} \;\gtrsim\; \underbrace{\frac{\alpha\eta^2}{\lambda_{\min}(H)}}_{\text{policy-signal variance}}\,, \tag{8}$$

equivalently $\mathrm{tr}(J_\pi \Sigma_\pi J_\pi^\top) \gtrsim \eta^2$ after the common $\alpha/\lambda_{\min}(H)$ factor cancels. Writing $J_\pi = \lambda \tilde{J}_\pi$ makes the λ-scaling explicit: $\lambda^2 \mathrm{tr}(\tilde{J}_\pi \Sigma_\pi \tilde{J}_\pi^\top) \gtrsim \eta^2$. This yields three testable predictions: **(a) Stationarity null**: $\Sigma_\pi \approx 0$ (supervised labels, symmetric slow drift) $\Rightarrow$ no pathology. **(b) Pathway null**: interventions that drive $J_\pi \to 0$ in the encoder subspace (stop-gradient, critic-side aux, annealed λ) $\Rightarrow$ recovery. **(c) Target ordering**: ranking target sources by the $J_\pi \cdot \Sigma_\pi$ coupling (random targets $\Rightarrow J_\pi{=}0$; frozen targets $\Rightarrow \Sigma_\pi{=}0$; co-learning $\Rightarrow$ both non-zero) predicts the pathology-severity ordering. All three are tested in §6.2–6.4.

Interpreting J_π. J_π is the sensitivity of the aux gradient to the target-policy action distribution; its column space coincides with the pathway through attention/GRU/ϕ_θ. It is large when the aux head is tightly tuned to the current target and small when the encoder is structurally insensitive (mean pooling, stop-gradient, or critic-side placement).

From cosine to proxy. Linearizing $\cos(g_\pi, g_{\text{aux}})$ at fixed norms, $\mathrm{Var}_t[\cos] \approx \hat{g}_\pi^\top \Sigma_\varepsilon \hat{g}_\pi / \|\bar{g}_{\text{aux}}\|^2 \leq \mathrm{tr}(\Sigma_\varepsilon)/\|\bar{g}_{\text{aux}}\|^2$, so the cosine temporal std is a finite-sample proxy for normalized $\mathrm{tr}(J_\pi \Sigma_\pi J_\pi^\top)^{1/2}$, measured on the final 50% window. The fixed-norm linearization can leak magnitude fluctuations into the angular metric; we rely on the empirical observation $\mathbb{E}_t[\cos] \approx 0$ (Figure 2b) to keep the magnitude contribution small. GAC's consecutive-step cosine diagnostic in async LLM RL [55] is a related but distinct temporal metric (single-stream autocorrelation, not policy-vs-aux alignment). Figure 5 shows $R^2 = 0.94$ across our four logged conditions (three have proxy zero by construction, so the plot verifies ordering more than slope). Within Full's 5 seeds, per-seed cosine std correlates with per-seed peak-to-Final50 drop at $r = +0.40$ (weak at n=5).

Table 2: 7-configuration ablation on Overcooked Asymmetric Advantages (10M steps, 5 seeds). Group A: 2×2 ablation over attention and auxiliary loss. Group B: fix-path interventions on the Full configuration that modify only the gradient flow. Best is comparable across all configurations (482–485, within ∼2 points); differentiation is purely in Final50 (late-training stability).

Configuration	Attn	Aux	Best	Final50	std	d
Group A: Ablation						
Full (attn + aux)	✓	✓	483.8	463.21	**8.55**	–
Neither (mean only)	–	–	482.6	469.70	3.72	0.88
No Attention (mean + aux)	–	✓	483.8	471.76	1.86	1.24
No Aux (attn only)	✓	–	484.4	**473.45**	3.62	**1.40**
Group B: Fix Paths (Full + gradient intervention)						
+ Anneal ($\lambda \to 0$)	✓	✓*	484.4	473.66	3.59	1.43
+ Stop-gradient	✓	✓†	482.6	471.60	**2.13**	1.20
+ Both	✓	✓*†	484.4	472.35	2.21	1.31

Cohen's d is vs. Full. *Aux λ linearly decayed $0.05 \to 0$ over first 50% of training. † `stop_gradient` on belief into aux head.

5 Methodology

5.1 Ablation Matrix and Fix-Path Interventions

We run VABL in a 2×2 ablation over attention vs. mean pooling and auxiliary loss on ($\lambda = 0.05$ constant) vs. off, yielding **Full** (attn+aux), **No Attention**, **No Aux**, and **Neither**. Three gradient-only interventions on Full: **Anneal** (linearly decay λ from 0.05 to 0 over the first 50% of training), **Stop-gradient** (`stop_gradient` on b_t^i before the aux head; aux MLP still trains but gradients do not reach the encoder), and **Anneal + stop-gradient** (both combined).

5.2 Target-Source Distinguishing Test

To distinguish directional interference from capacity consumption, we run Full VABL with aux network, capacity, and gradient pathway identical to Full, varying *only* the target source: **Frozen-policy targets** (argmax of an iteration-0 random-init snapshot π_{θ_0}: stationary but structured) and **Uniform random targets** (resampled each iteration: stationary and unstructured). Capacity consumption predicts both to be pathological; directional interference predicts recovery scaled by residual structure.

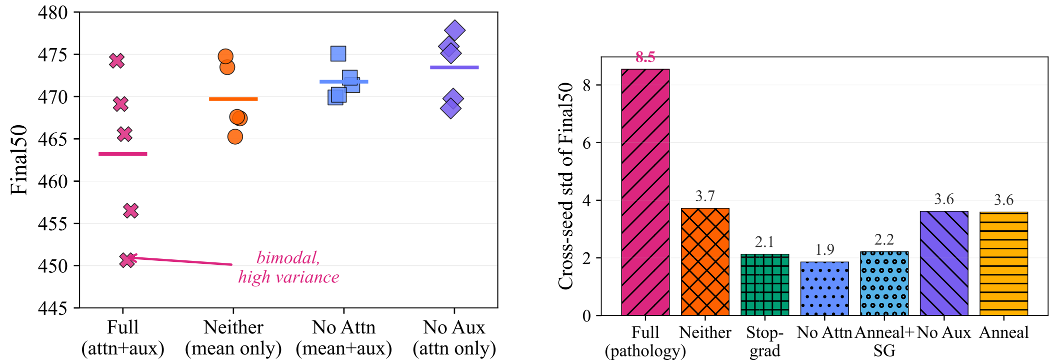
5.3 Environments, baselines, and metrics

Four JaxMARL-based [41] cooperative MARL environments (Overcooked [6] Asymmetric Advantages and Cramped Room, Hanabi 2-player [2], MPE simple_spread [31], SMAX 3v3 [42, 15]) plus CIFAR-100 with a small ViT + RotNet aux head as a supervised-stationary control. Baselines cover three paradigms (decentralized policy gradient: MAPPO [58]; communication: TarMAC [12], CommNet [46]; attention: AERIAL [39]), reimplemented in JAX with matched hyperparameters. Metrics are Best (peak episode reward or test accuracy), Final50 (mean of last 50 episodes; Final5 for vision), cross-seed std, and Cohen's d [11]; mean ± std over 5 seeds throughout. Full environment specs, baseline description, and metric definitions in Appendix J.

6 Results

6.1 Isolating the Pathology in VABL

Only Full exhibits the pathology: Cohen's $d \geq 0.88$ vs every alternative, with cross-seed std 2.3–4.6× larger, while all seven configurations reach comparable Best (within ∼2 points). The 2×2 interaction is clean on AA: the instability requires *both* attention and the auxiliary loss; the Neither vs No Attention gap is within one std (Appendix E). All three fix paths (Group B) land in Final50 $[471.60, 473.66]$, indistinguishable from No Aux; stop-gradient alone drops std $8.55 \to 2.13$. The aux-to-encoder pathway is thus necessary, but this does not yet distinguish directional interference from generic capacity consumption.



(a) Per-seed Final50.

(b) Cross-seed std across 7 configurations.

Figure 3: Cross-seed dispersion on Overcooked AA (10M steps, 5 seeds). **(a)** Full VABL (magenta) has cross-seed std of 8.55 (2.3–4.6$\times$ larger than alternatives); individual Full seeds drift 8–34 points below their own peaks (per-seed Best minus Final50); horizontal lines show means. **(b)** Std drops sharply once the aux-to-encoder gradient is cut: Full (8.55) $\rightarrow$ Anneal (3.59) $\rightarrow$ Stop-grad (2.13); No Aux (3.62) removes the conduit entirely. Marker shapes and hatching vary for grayscale readability.

Table 3: Target-source distinguishing test on Overcooked AA (10M steps, 5 seeds). Aux network and gradient pathway identical to Full; only the target source varies. Teammate co-learning targets are pathological (Cohen's $d = +1.40$ vs No Aux, bootstrap CI $[+0.47, +3.58]$); frozen and random targets are both not resolved from No Aux at n=5 (both CIs cross zero, §6.2; absence of a resolved effect, not equivalence), with random targets producing the tightest cross-seed std in the paper.

Target source	**Structured?**	**Non-stationary?**	**Final50**	**std**
Teammate co-learning (Full)	yes	yes	463.21	**8.55**
Frozen initial-policy	yes	no	470.22	4.04
Uniform random	no	no	**473.58**	**1.66**
No aux (reference)	n/a	n/a	473.45	3.62

6.2 Causal Identification of the Mechanism

The fix paths are consistent with two distinct stories: directional interference driven by structured non-stationary aux targets, and a generic aux-capacity-consumption alternative in which any aux head attached to the shared encoder degrades late-training performance regardless of what the aux gradient encodes. Two independent controls are inconsistent with the latter.

Target-source test. We hold aux capacity, architecture, and the aux-to-encoder gradient pathway identical to Full and vary only the *source* of the aux target (§5.2). The capacity-consumption hypothesis predicts all three conditions to be pathological. Table 3 is inconsistent with this: at n=5 teammate co-learning remains distinct from No Aux (d= $+1.40$, CI $[+0.47, +3.58]$),[1] while both frozen-vs-No Aux (d= $+0.75$, CI $[-0.44, +2.73]$) and random-vs-No Aux (d= $+0.04$, CI $[-1.40, +1.47]$) have CIs that cross zero (no resolved effect, not a formal equivalence test). Structured co-learning targets carry teammate-policy-specific information that drifts across iterations; frozen targets remove the drift, random targets remove both. Only the structured-and-drifting combination degrades Final50 stability.

Aux-capacity scaling control. As a complementary test of the same alternative, we vary the aux-MLP hidden dimension $\{16, 32, 128\}$ against the canonical 64 with teammate targets. A capacity-consumption hypothesis predicts monotonic worsening as hidden dim grows; the directional-interference story predicts no effect.

All four Cohen's d bootstrap CIs cross zero; no capacity level is distinguishable from 64 (Final50 means in $[462.89, 468.74]$, cross-seed stds in $[4.91, 8.55]$). An $8\times$ sweep produces no resolvable effect, reinforcing the target-source conclusion: capacity consumption alone cannot explain the

[1]Throughout we use $d = (\text{No Aux} - X)/\sigma_{\text{pooled}}$, so positive means No Aux is better.

Table 4: Aux-capacity scaling on Overcooked AA (10M steps, 5 seeds). Full VABL with teammate targets, varying only aux-MLP hidden dim. Final50 and std are effectively flat across 8× capacity range.

aux hidden dim	**Final50**	**std**	d **vs 64**
16	462.89	8.18	$+0.03$
32	468.74	4.91	-0.71
64 (canonical)	$\mathbf{463.21}$	$\mathbf{8.55}$	n/a
128	466.35	7.75	-0.34

Table 5: Cross-environment generalization. Cohen's d is No Aux vs Full on each task; positive means aux hurts. Final50 for MARL, Final5 for CIFAR-100; all std values are the population estimator (consistent with Table 2); 5 seeds per cell.

Environment	**Type**	**Full**	**No Aux**	d
Overcooked Cramped Room	cooperative	409.70 ± 1.26	416.18 ± 0.71	$+\mathbf{5.07}$
Hanabi (2p, turn-based)	cooperative	0.14 ± 0.23	2.81 ± 1.18	$+2.81$
Overcooked Asymm. Advantages	cooperative	463.21 ± 8.55	473.45 ± 3.62	$+1.40$
SMAX 3v3	cooperative	10.60 ± 0.41	11.15 ± 0.37	$+1.26$
MPE simple_spread	cooperative	-18.82 ± 0.89	-18.77 ± 0.91	~ 0
CIFAR-100 (ViT)	supervised	42.78 ± 0.27	43.18 ± 0.37	$+1.09$

Final50 instability. Together with the target-source test, this is inconsistent with capacity-consumption as the driver on two independent axes.

6.3 Fix Paths, Given the Mechanism

Each Group-B intervention drives $J_\pi \to 0$ in the encoder subspace: λ-annealing decays the aux contribution, stop-gradient severs it at the belief, critic-side placement routes it away from the actor encoder. All land in Final50 $[471.60, 473.66]$, indistinguishable from No Aux (Appendix E).

6.4 Cross-Environment Generalization

The pathology reproduces on all four cooperative-coordination benchmarks ($d \in [+1.26, +5.07]$) and is absent on MPE and supervised CIFAR-100. The large $d = +5.07$ on Cramped Room reflects that environment's unusually tight cross-seed std (~ 1) rather than a larger mean gap (~ 6 points); Hanabi's absolute scores are low, so the relative ordering rather than the magnitude is the claim (Appendix E). On Hanabi the 2×2 loses AA's attention specificity: both aux-ON cells are pathological, both aux-OFF healthy, consistent with partial information making even mean-pool aux disruptive. As teammate drift and coordination difficulty grow, more pathways become vulnerable; in stationary supervised learning the drift is zero and the aux is near-null ($d = +1.09$ on CIFAR-100, ~ 0.4 percentage points absolute, two orders of magnitude smaller than MARL Best→Final50 drops; boundary-case discussion in Appendix D). Proposition 1 formalizes this scaling.

On SMAX, fix paths reduce cross-seed std but not the mean; halving λ worsens Final50 (10.18 ± 0.35 vs 10.60 at λ=0.05), a non-monotonic response we flag as open (Appendix E).

6.5 Generalization across architectures and λ

A 60-run architecture sweep and a 3-point λ sweep confirm generalization: the pathology reproduces with LSTM + cross-attention (d=0.80) and GRU + additive attention (d=1.01), is absent under mean pooling ($d = -0.30$), and the critic-side aux configuration moves to a qualitatively different regime ($d = -0.79$ on the mean, with cross-seed std much larger). At 50M steps the λ sweep retains cross-seed std > 7 at both λ=0.01 and λ=0.05 (Appendices I, H).

7 Limitations

- **First-order model.** Local quadratic loss, i.i.d. noise, linear expansion in π_j; J_π and Σ_π are not directly observed but proxied by the normalized cosine temporal std (Figure 5). The bound is tightest in late training; autocorrelated $\varepsilon(t)$ inflates the right-hand side by an $O(\tau_c)$ factor without changing the functional form (Appendix A).
- **λ-scaling is local.** The λ^2 prediction holds within a λ range where aux-head representations are qualitatively unchanged; our 3-point sweep is non-monotonic because J_π co-adapts with λ at small λ (Appendix H).
- **Out-of-scope patterns.** The critic-side aux configuration ($d = -0.79$ on the mean, Appendix I) is a qualitatively different regime (higher mean but much higher cross-seed std), not improved stability; on SMAX, fix paths recover variance but not the mean gap, inconsistent with a purely zero-mean noise mechanism.
- **Empirical scale.** 5 seeds per condition on all MARL cells and CIFAR-100; the architecture sweep resolves at positive d in 2 of 4 configurations; absolute Hanabi scores are low despite clean relative ordering.
- **Future work.** Eq. (8) suggests a drift-gated controller that toggles `stop_gradient` on the aux-to-encoder pathway when a sliding-window estimate of $\mathrm{std}(\cos)$ exceeds a threshold; DG-PG [57] (Theorem 4.2) gives the closed-form optimum it approximates, and COALA-PG [33] formalizes the co-learning meta-POMDP underlying Σ_π. Empirical evaluation of the schedule and of replication on BEPAL [22] and Dynamic Belief [60] is immediate future work.

8 Conclusion

A constant auxiliary weight on a target that is both task-coupled and co-adaptive is not a free regularizer; it is a silent late-training stability risk whose severity scales with how fast the target drifts relative to the vanishing policy gradient. When either ingredient is absent, the aux is harmless.

More broadly, gradient conflict in deep policy learning is not always a magnitude problem and not always a multi-task problem. Directional, target-driven interference is a distinct failure mode with its own diagnostic (temporal gradient-cosine variance) and its own remedy (sever the encoder-side pathway). A drift-gated aux controller approximating the DG-PG [57] closed-form weight is the natural successor; the critic-side reversal and SMAX residual mean gap remain open.

References

[1] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. Neural machine translation by jointly learning to align and translate. In *International Conference on Learning Representations*, 2015.

[2] Nolan Bard, Jakob N Foerster, Sarath Chandar, Neil Burch, Marc Lanctot, H Francis Song, Emilio Parisotto, Vincent Dumoulin, Subhodeep Moitra, Edward Hughes, Iain Dunning, Shibl Mourad, Hugo Larochelle, Marc G Bellemare, and Michael Bowling. The Hanabi challenge: A new frontier for AI research. *Artificial Intelligence*, 280:103216, 2020.

[3] Matteo Bettini, Amanda Prorok, and Vincent Moens. BenchMARL: Benchmarking multi-agent reinforcement learning. *Journal of Machine Learning Research*, 25(217):1–10, 2024.

[4] James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas, Skye Wanderman-Milne, and Qiao Zhang. JAX: Composable transformations of Python+NumPy programs. `http://github.com/google/jax`, 2018.

[5] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *International Conference on Computer Vision*, 2021.

[6] Micah Carroll, Rohin Shah, Mark K Ho, Tom Griffiths, Sanjit Seshia, Pieter Abbeel, and Anca Dragan. On the utility of learning about humans for human-AI coordination. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[7] Xinlei Chen and Kaiming He. Exploring simple Siamese representation learning. In *Conference on Computer Vision and Pattern Recognition*, 2021.

[8] Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *International Conference on Machine Learning*. PMLR, 2018.

[9] Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Conference on Empirical Methods in Natural Language Processing*, 2014.

[10] Raymond Chua, Arna Ghosh, Christos Kaplanis, Blake A. Richards, and Doina Precup. Learning successor features the simple way. In *Advances in Neural Information Processing Systems*, 2024.

[11] Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, 2 edition, 1988.

[12] Abhishek Das, Théophile Gervet, Joshua Romoff, Dhruv Batra, Devi Parikh, Mike Rabbat, and Joelle Pineau. TarMAC: Targeted multi-agent communication. In *International Conference on Machine Learning*. PMLR, 2019.

[13] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

[14] Yunshu Du, Wojciech M Czarnecki, Siddhant M Jayakumar, Mehrdad Farajtabar, Razvan Pascanu, and Balaji Lakshminarayanan. Adapting auxiliary losses using gradient similarity. *arXiv preprint arXiv:1812.02224*, 2018.

[15] Benjamin Ellis, Jonathan Cook, Skander Moalla, Mikayel Samvelyan, Mingfei Sun, Anuj Mahajan, Jakob Foerster, and Shimon Whiteson. SMACv2: An improved benchmark for cooperative multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023.

[16] Ching Fang and Kimberly L. Stachenfeld. Predictive auxiliary objectives in deep reinforcement learning mimic learning in the brain. In *International Conference on Learning Representations*, 2024.

[17] Claude Formanek, Callum Rhys Tilbury, Louise Beyers, Jonathan P. Shock, and Arnu Pretorius. Dispelling the mirage of progress in offline MARL through standardised baselines and evaluation. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024.

[18] Spyros Gidaris, Praveer Singh, and Nikos Komodakis. Unsupervised representation learning by predicting image rotations. In *International Conference on Learning Representations*, 2018.

[19] Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pinto, Zhan Zheng, Mohammad Azizi, et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[20] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023.

[21] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

[22] Qinwei Huang, Stefan Wang, Simon Khan, Garrett E. Katz, and Qinru Qiu. Predictive auxiliary learning for belief-based multi-agent systems. *arXiv preprint arXiv:2511.01078*, 2025.

[23] Shariq Iqbal and Fei Sha. Actor-attention-critic for multi-agent reinforcement learning. In *International Conference on Machine Learning*. PMLR, 2019.

[24] Max Jaderberg, Volodymyr Mnih, Wojciech Marian Czarnecki, Tom Schaul, Joel Z Leibo, David Silver, and Koray Kavukcuoglu. Reinforcement learning with unsupervised auxiliary tasks. *arXiv preprint arXiv:1611.05397*, 2016.

[25] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

[26] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

[27] Michael Laskin, Aravind Srinivas, and Pieter Abbeel. CURL: Contrastive unsupervised representations for reinforcement learning. In *International Conference on Machine Learning*. PMLR, 2020.

[28] Wenhao Li, Xiangfeng Wang, Bo Jin, Junjie Sheng, and Hongyuan Zha. Dealing with non-stationarity in MARL via trust-region decomposition. In *International Conference on Learning Representations*, 2022.

[29] Xingyu Lin, Harjatin Singh Baweja, George Kantor, and David Held. Adaptive auxiliary task weighting for reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[30] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

[31] Ryan Lowe, Yi I Wu, Aviv Tamar, Jean Harb, OpenAI Pieter Abbeel, and Igor Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[32] Clare Lyle, Mark Rowland, Georg Ostrovski, and Will Dabney. On the effect of auxiliary tasks on representation dynamics. In *International Conference on Artificial Intelligence and Statistics*. PMLR, 2021.

[33] Alexander Meulemans, Seijin Kobayashi, Johannes von Oswald, Nino Scherrer, Eric Elmoznino, Blake Richards, Guillaume Lajoie, Blaise Agüera y Arcas, and João Sacramento. Multi-agent cooperation through learning-aware policy gradients. *arXiv preprint arXiv:2410.18636*, 2024.

[34] Skander Moalla, Andrea Miele, Daniil Pyatko, Razvan Pascanu, and Caglar Gulcehre. No representation, no trust: Connecting representation, collapse, and trust issues in PPO. In *Advances in Neural Information Processing Systems*, 2024.

[35] Hadi Nekoei, Akilesh Badrinaaraayanan, Amit Sinha, Mohammad Amini, Janarthanan Rajendran, Aditya Mahajan, and Sarath Chandar. Dealing with non-stationarity in decentralized cooperative multi-agent deep reinforcement learning via multi-timescale learning. In *Conference on Lifelong Learning Agents*. PMLR, 2023.

[36] Frans A Oliehoek and Christopher Amato. *A Concise Introduction to Decentralized POMDPs*. Springer, 2016.

[37] George Papadopoulos, Andreas Kontogiannis, Foteini Papadopoulou, Chaido Poulianou, Ioannis Koumoutsos, and George Vouros. An extended benchmarking of multi-agent reinforcement learning algorithms in complex fully cooperative tasks. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2025.

[38] Santosh Patapati. Gradient interference-aware graph coloring for multi-task learning. *arXiv preprint arXiv:2509.16959*, 2025.

[39] Thomy Phan, Lenz Belzner, Thomas Gabor, Andreas Sedlmeier, Fabian Ritz, and Claudia Linnhoff-Popien. Attention-based recurrence for multi-agent reinforcement learning under stochastic partial observability. In *International Conference on Machine Learning*. PMLR, 2023.

[40] Tabish Rashid, Mikayel Samvelyan, Christian Schroeder De Witt, Gregory Farquhar, Jakob Foerster, and Shimon Whiteson. Monotonic value function factorisation for deep multi-agent reinforcement learning. *Journal of Machine Learning Research*, 21(178):1–51, 2020.

[41] Alexander Rutherford, Benjamin Ellis, Matteo Gallici, Jonathan Cook, Andrei Lupu, Gardar Ingvarsson, Timon Willi, Ravi Hammond, Akbir Khan, Christian Schroeder de Witt, and Jakob Foerster. JaxMARL: Multi-agent RL environments and algorithms in JAX. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024.

[42] Mikayel Samvelyan, Tabish Rashid, Christian Schroeder de Witt, Gregory Farquhar, Natasha Nardelli, Tim GJ Rudner, Chia-Man Hung, Philip HS Torr, Jakob Foerster, and Shimon Whiteson. The StarCraft multi-agent challenge. In *International Conference on Autonomous Agents and Multiagent Systems*, 2019.

[43] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations*, 2016.

[44] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[45] Max Schwarzer, Ankesh Anand, Rishab Garg, R Devon Graves, Ian Goodfellow, and Marc G Bellemare. Data-efficient reinforcement learning with self-predictive representations. In *International Conference on Learning Representations*, 2021.

[46] Sainbayar Sukhbaatar, Arthur Szlam, and Rob Fergus. Learning multiagent communication with backpropagation. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

[47] Guoheng Sun, Tingting Du, Kaixi Feng, Chenxiang Luo, Xingguo Ding, Zheyu Shen, Ziyao Wang, Yexiao He, and Ang Li. ROCKET: Residual-oriented multi-layer alignment for spatially-aware vision-language-action models. *arXiv preprint arXiv:2602.17951*, 2026.

[48] Kale-ab Abebe Tessera, Arrasy Rahman, Amos Storkey, and Stefano V Albrecht. HyperMARL: Adaptive hypernetworks for multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems*, 2025.

[49] Callum Rhys Tilbury, Claude Formanek, Louise Beyers, Jonathan P. Shock, and Arnu Pretorius. Coordination failure in cooperative offline MARL. In *ICML 2024 Workshop on Aligning Reinforcement Learning Experimentalists and Theorists*, 2024.

[50] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[51] Claas A. Voelcker, Tyler Kastner, Igor Gilitschenski, and Amir-massoud Farahmand. When does self-prediction help? understanding auxiliary tasks in reinforcement learning. In *Reinforcement Learning Conference (RLC)*, 2024.

[52] Muning Wen, Jakub Grudzien Kuba, Runji Lin, Weinan Zhang, Ying Wen, Jun Wang, and Yaodong Yang. Multi-agent reinforcement learning is a sequence modeling problem. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

[53] Yi-Ning Weng and Hsuan-Wei Lee. How exploration breaks cooperation in shared-policy multi-agent reinforcement learning. *arXiv preprint arXiv:2601.05509*, 2026.

[54] Peiyao Xiao, Hao Ban, and Kaiyi Ji. Direction-oriented multi-objective learning: Simple and provable stochastic algorithms. *Advances in Neural Information Processing Systems*, 2023.

[55] Haofeng Xu, Junwei Su, Yukun Tian, Lansong Diao, Zhengping Qian, and Chuan Wu. GAC: Stabilizing asynchronous RL training for LLMs via gradient alignment control. *arXiv preprint arXiv:2603.01501*, 2026.

[56] Jianye Xu, Omar Sobhy, and Bassam Alrifaee. XP-MARL: Auxiliary prioritization in multi-agent reinforcement learning to address non-stationarity. *arXiv preprint arXiv:2409.11852*, 2024.

[57] Shan Yang and Yang Liu. Descent-guided policy gradient for scalable cooperative multi-agent learning. *arXiv preprint arXiv:2602.20078*, 2026.

[58] Chao Yu, Akash Velu, Eugene Vinitsky, Jiaxuan Gao, Yu Wang, Alexandre Bayen, and Yi Wu. The surprising effectiveness of PPO in cooperative multi-agent games. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, volume 35, 2022.

[59] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[60] Yunpeng Zhai, Peixi Peng, Chen Su, and Yonghong Tian. Dynamic belief for decentralized multi-agent cooperative learning. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence (IJCAI)*, pages 344–352, 2023.

[61] Qi Zhang, Peiyao Xiao, Shaofeng Zou, and Kaiyi Ji. MGDA converges under generalized smoothness, provably. In *International Conference on Learning Representations*, 2025.

A Extended Theory Discussion

This appendix expands the compressed theory exposition in §4. It is the long-form version of material referenced but not stated in full in the main text, organized as follows. §A.1 specifies exactly when the first-order linearization $\varepsilon(t) \approx J_\pi(\pi_j^{(t)} - \bar{\pi}_j)$ is accurate, and why restricting the analysis window to the final 50% of training is the correct regime for measuring the cosine-std proxy. §A.2 extends Proposition 1 from the i.i.d. noise assumption used in the main body to colored noise with integrated autocorrelation time τ_c, showing that τ_c enters as a multiplicative factor on the bound without changing its functional form. §A.3 states precisely why the λ^2 variance scaling is a local rather than a global prediction, and documents the measured non-monotonicity of the 3-point λ sweep. §A.4 unpacks what J_π represents in a VABL architecture: the encoder-subspace column through attention/GRU/ϕ_θ by which teammate one-hot inputs influence aux-head predictions, and why each fix path (mean pooling, stop-gradient, critic-side placement) drives this projection toward zero. §A.5 collects the six scope limits and open questions at the end of the main-body theory section in fully expanded form, including the SMAX residual mean gap, the critic-side reversal, and the proxy-plot categorical structure. §A.6 places our theory objects alongside five recent gradient-interference and non-stationarity formalisms (COALA-PG, DG-PG, ROCKET, GAC, Trust-Region Decomposition) and states precisely which formal assumptions our setting violates relative to each.

A.1 Linearization, locality, and the analysis window

The first-order expansion $\varepsilon(t) \approx J_\pi(\pi_j^{(t)} - \bar{\pi}_j)$ is exact when the target-generating policy stays within a small neighborhood of the time-averaged policy $\bar{\pi}_j$ over the analysis window. Formally, write $g_{\mathrm{aux}}(t) = g_{\mathrm{aux}}(\bar{\pi}_j) + J_\pi \Delta_t + \mathcal{O}(\|\Delta_t\|^2)$ with $\Delta_t = \pi_j^{(t)} - \bar{\pi}_j$; the drop-to-noise factorization $\Sigma_\varepsilon \approx J_\pi \Sigma_\pi J_\pi^\top$ is accurate to $\mathcal{O}(\|\Delta\|^3)$ in the window's maximal drift. Restricting to the final 50% of training is important because policy convergence implies smaller per-iteration drift, so the linearization is tightest precisely where we measure the cosine-std proxy. Mid-training, where the worst cross-seed instability actually occurs, the linearization is less tight, so the theory's quantitative predictions are most useful for late-training equilibrium behavior and qualitative for mid-training transients.

A.2 Colored-noise correction to Proposition 1

Proposition 1 treats $\varepsilon(t)$ as i.i.d. to obtain a clean closed-form bound. Real $\varepsilon(t)$ is autocorrelated because consecutive target-generating policies differ by a single gradient step, so $\pi_j^{(t+1)}$ and $\pi_j^{(t)}$ are close in parameter space. Let $\tau_c = \sum_{k=-\infty}^{\infty} \rho_\varepsilon(k)$ denote the (two-sided) integrated autocorrelation time of ε, where $\rho_\varepsilon(k) = \mathrm{tr}(\mathrm{Cov}[\varepsilon(t), \varepsilon(t+k)])/\mathrm{tr}(\mathrm{Var}[\varepsilon(t)])$ is the trace-normalized scalar autocorrelation function. For an AR(1) approximation with decay rate $\phi \in [0, 1)$, $\tau_c = (1+\phi)/(1-\phi)$. The steady-state bound becomes

$$\mathbb{E}[\|\delta\|^2]_{\mathrm{colored}} \;\propto\; \tau_c \cdot \frac{\alpha\, \mathrm{tr}(J_\pi\, \Sigma_\pi\, J_\pi^\top)}{\lambda_{\min}(H)}. \tag{9}$$

The dependence on $\mathrm{tr}(J_\pi \Sigma_\pi J_\pi^\top)$ and $\lambda_{\min}(H)$ is unchanged; τ_c enters as a multiplicative factor that can be large for fast-learning teammates (AR(1) with ϕ close to 1). This means the i.i.d. bound may underestimate absolute variance substantially, but the *ordering* across conditions is preserved under any uniform τ_c.

A.3 Scope of the λ prediction

Since $g_{\mathrm{aux}} \propto \lambda$ by construction, the first-order bound predicts displacement variance $\propto \lambda^2$. Our 3-point λ sweep (Appendix H) is *not* monotonic: Final50 std at $\lambda \in \{0.001, 0.01, 0.05\}$ is $\{6.37, 1.67, 8.55\}$ respectively. The bound holds locally (variance increases from $\lambda = 0.01$ to $\lambda = 0.05$) but fails across the full range because aux-head representations co-adapt with λ: at very small λ the aux head receives too little signal to organize its representation, so J_π itself becomes a function of λ in a way the first-order analysis does not capture. The prediction should be read as "within a λ range where the aux-head representation is qualitatively unchanged," not as a global claim.

A.4 Interpreting J_π in VABL

J_π is the sensitivity of the aux gradient to the target-policy action distribution, evaluated around $\bar{\pi}_j$. In VABL with teammate-action targets, its column space coincides with the pathway through attention/GRU/ϕ_θ by which teammate one-hot inputs influence aux-head predictions. It is large when the aux head is tightly tuned to the current target, and small when the encoder is structurally insensitive: mean pooling collapses heterogeneous target sources onto a single mean vector, so different target identities produce identical contexts; stop-gradient on b_t^i cuts the encoder-side coupling while leaving the aux MLP trainable; critic-side placement routes g_{aux} through the critic parameters entirely, so the actor encoder never receives the aux signal. All three interventions drive the *encoder-subspace* projection of J_π toward zero without eliminating the aux loss itself.

A.5 Open questions and scope limits

The main text compresses six caveats to the first-order model; this section states each in full. **(i) Autocorrelation.** The i.i.d. noise assumption gives the variance-scaling *shape* but not absolute magnitude; colored-noise correction (Appendix A.2) multiplies the bound by τ_c. **(ii) Local-quadratic approximation.** The bound is tightest in late training where drift is smallest, which is also where the pathology is least severe; it applies less tightly mid-training where the worst cross-seed instability actually occurs. **(iii) Proxy–variance ordering.** The $R^2 = 0.94$ fit (Figure 5) is dominated by the categorical separation between Full (non-zero proxy) and the three pathway-zero conditions. The plot verifies ordering more than slope. **(iv) Critic-side reversal.** Routing g_{aux} through the critic produces a qualitatively different regime: the mean Final50 increases ($d = -0.79$ in the No-Aux-vs-Full sense, Appendix I) but the cross-seed std also increases substantially. This is not ”improved stability” under our variance-based definition; it is a different operating point entirely (higher mean, much higher variance) that Proposition 1 does not describe. **(v) SMAX residual mean gap.** Fix paths recover variance but not mean on SMAX; this is inconsistent with a purely zero-mean directional-noise mechanism (symmetric oscillation should not bias the mean). Either a secondary pathology coexists in SMAX or the zero-mean assumption of $\varepsilon(t)$ is too strong there. **(vi) λ non-monotonicity.** The monotonic-λ prediction fails globally because aux-head representations co-adapt with λ, making J_π itself λ-dependent (Appendix A.3).

A.6 Relation to other gradient-interference formalisms

Our theory sits at a specific axis within a broader literature of gradient-interference and non-stationarity diagnostics; related formal objects are listed below.

DG-PG exogeneity assumption [57]. Their framework treats a guidance term G with reference trajectory $\tilde{x}_t$ satisfying $\nabla_\theta \tilde{x}_t = 0$ (Assumption 3.1) as a variance-reduction signal for cooperative MARL. Their Theorem 4.2 then gives the closed-form optimal combining weight

$$\min_{\alpha \in (0,1)} \operatorname{Var}(\hat{g}^i_{\text{DG}}) = \frac{\sigma_J^2 \sigma_G^2 (1 - \rho^2)}{\sigma_J^2 + \sigma_G^2 + 2\rho\, \sigma_J \sigma_G}, \tag{10}$$

where σ_J^2, σ_G^2 are the variances of the policy-gradient and guidance-gradient estimators and $\rho \in (-1, 0)$ is their correlation under DG-PG's sign convention (the $+2\rho$ in the denominator becomes $-2|\rho|$ since ρ is negative by their Assumption 3.2). In our setting the aux target is generated by a co-evolving policy (the teammate's in Full VABL), so $\nabla_\theta \tilde{x}_t \neq 0$. Assumption 3.1 is violated, which is precisely why Σ_ε is driven by Σ_π rather than collapsing. Reading $\hat{g}_J = g_\pi$, $\hat{g}_G = g_{\text{aux}}$, $\rho = \mathbb{E}_t[\cos]$, the DG-PG formula prescribes an optimal $\alpha^*(t)$ that our drift-gated aux controller approximates by binary gating.

COALA-PG batched meta-POMDP [33]. Meulemans et al. formalize cooperative-MARL learning as a batched co-player shaping POMDP whose meta-state is the joint parameter vector (ϕ^i, ϕ^{-i}). The distribution over meta-trajectories induced by the meta-dynamics is the rigorous object that our Σ_π informally approximates.

Trust-Region Decomposition [28]. Li et al. introduce δ-stationarity, an explicit non-stationarity measure bounded by the KL-divergence of consecutive joint policies. When we say “target-policy drift,” this is the formal quantity being measured; the KL bound gives an upper estimate on $\operatorname{tr}(\Sigma_\pi)$ in natural parameter coordinates.

ROCKET cross-layer interference [47]. Sun et al. analyze gradient interference in multi-layer representation alignment, decomposing the squared gradient norm as $\|G\|^2 = \sum_l \alpha_l^2 \|v_l\|^2 + \sum_{a \neq b} \alpha_a \alpha_b \langle v_a, v_b \rangle$, where $\langle v_a, v_b \rangle = g_a^\top M_{ab} g_b$ is a bilinear form on teacher-error signals. Their cross-term is the static (cross-layer) analog of our temporal cross-term: both identify $\langle g_a, g_b \rangle$ structure as the source of conflict, and both propose structural constraints (shared projector for ROCKET, stop-gradient for us) that eliminate the conflict without weight tuning.

GAC consecutive-step cosine [55]. Xu et al. diagnose asynchronous LLM RL instability via the consecutive-step cosine $c_t = \langle g_t, g_{t-1} \rangle / (\|g_t\| \|g_{t-1}\|)$ of a single gradient stream. This is a temporal autocorrelation-at-lag-1; our $\cos(g_\pi, g_{\mathrm{aux}})$ is a cross-correlation at lag-0. Both are cosine-based first-order diagnostics and share the "late-window temporal std" estimation pattern, but diagnose different instabilities: theirs is staleness across async workers, ours is structured-target interference within a single worker.

B Methodology Details

This appendix expands the compressed methodology descriptions in §5, preserving specifics trimmed for the page limit. §B.1 gives the full specification of the target-source distinguishing test, including the procedure used to construct frozen-policy targets from an iteration-0 random-init snapshot, the uniform-random resampling protocol, and the predictions each target source makes under the capacity-consumption versus directional-interference hypotheses. §B.2 describes each baseline (MAPPO, TarMAC, CommNet, AERIAL) with the paradigm it exemplifies, documents the shared JAX/JaxMARL reimplementation and hyperparameter matching that make cross-method differences interpretable, and explains why these baselines do not serve as controls for the core mechanism (no baseline instantiates the attention + teammate-prediction-aux design pattern).

B.1 Target-source distinguishing test: full specification

The test compares Full VABL with three alternative aux-target sources, holding aux network, capacity, and the aux-to-encoder gradient pathway identical across conditions:

- **Teammate co-learning (Full).** Standard VABL: the aux target for step $t + 1$ is the actual teammate action a_{t+1}^j. Targets are both *structured* (come from a real policy) and *non-stationary* (drift with teammate co-learning).
- **Frozen-policy targets.** At iteration 0 we snapshot the initial (random-init) agent parameters as θ_0 and freeze them. At each subsequent iteration, the aux target for step $t + 1$ is $\arg\max_a \pi_{\theta_0}(a \mid o_{t+1}^j)$ applied to the observed rollout state, rather than the teammate's actual action. Targets are *stationary* (no drift with teammate co-learning) but remain *structured* (from a real policy). Realizes the $\Sigma_\pi \to 0$ limit.
- **Uniform random targets.** The aux target for step $t + 1$ is an integer drawn uniformly at random from $\{0, \ldots, |\mathcal{A}| - 1\}$, resampled each iteration. Targets are *stationary in expectation* and *unstructured* (no co-learning signal, no semantic coupling to teammates). Realizes the $J_\pi \to 0$ limit.

All three interventions leave aux capacity, architecture, gradient pathway, and loss form identical to Full; they vary *only* the source and structure of the aux target. The capacity-consumption hypothesis predicts all three conditions to be pathological. The directional-interference hypothesis predicts a graded recovery: teammate-co-learning stays pathological, while frozen and random targets both move back toward the No Aux regime (at n=5 the frozen- and random-vs-No-Aux Cohen's d bootstrap CIs both cross zero; only the teammate-vs-No-Aux comparison resolves).

B.2 Baselines: full description and comparability rationale

We compare against four methods spanning three paradigms: **MAPPO** [58] (decentralized policy gradient, centralized critic), **TarMAC** [12] (targeted attention-based communication), **CommNet** [46] (broadcast communication), and **AERIAL** [39] (attention over teammate hidden states; our port uses PPO for comparability, the original is QMIX-based). All baselines are reimplemented in JAX [4] using JaxMARL [41] with matched hyperparameters, env wrapper, and evaluation protocol, so cross-method differences reflect algorithmic choices rather than infrastructure drift. None of these baselines exhibits the attention + teammate-prediction-aux design pattern, so they are reported for context rather than as controls for the core mechanism. Direct re-implementation of Dynamic Belief [60] or BEPAL [22], which do instantiate the pattern, would be a stronger external comparison and is left to future work; we rely instead on Zhai et al.'s independently-reported Static-Belief failure mode as external corroboration (§2).

C Full Proof of Proposition 1

Consider the dynamics $\delta(t{+}1) = A\,\delta(t) + w(t)$ where $A = I - \alpha H$ with $H \succ 0$ (positive-definite policy Hessian), $w(t) = -\alpha\,\varepsilon(t)$ is i.i.d. with $\mathbb{E}[w] = 0$ and $\mathrm{Cov}[w] = \alpha^2 \Sigma_\varepsilon$.

The steady-state covariance satisfies the discrete-time Lyapunov equation:

$$\Sigma_\delta = A\,\Sigma_\delta\,A^T + \alpha^2\,\Sigma_\varepsilon. \tag{11}$$

Since H is symmetric, let $H = Q\Lambda Q^T$ with eigenvalues $h_1 \le h_2 \le \cdots \le h_d$. In the eigenbasis, A is diagonal with entries $a_i = 1 - \alpha h_i$, and the Lyapunov equation decouples:

$$(\Sigma_\delta)_{ii} = \frac{\alpha^2\,(\Sigma_\varepsilon)_{ii}}{1 - a_i^2} = \frac{\alpha^2\,(\Sigma_\varepsilon)_{ii}}{2\alpha h_i - \alpha^2 h_i^2}. \tag{12}$$

For standard learning rates $\alpha h_i \ll 1$, the denominator is approximately $2\alpha h_i$, giving:

$$(\Sigma_\delta)_{ii} \approx \frac{\alpha\,(\Sigma_\varepsilon)_{ii}}{2h_i}. \tag{13}$$

The trace (total steady-state variance) is:

$$\mathbb{E}[\|\delta\|^2] = \mathrm{tr}(\Sigma_\delta) \approx \frac{\alpha}{2}\sum_i \frac{(\Sigma_\varepsilon)_{ii}}{h_i} \;\le\; \frac{\alpha\,\mathrm{tr}(\Sigma_\varepsilon)}{2\,h_{\min}}, \tag{14}$$

which establishes the bound in Proposition 1. The bound is tight when Σ_ε is concentrated on the smallest-eigenvalue eigenvector of H (the most weakly-restoring direction of the policy loss landscape); for isotropic noise it loosens by up to a factor of d because the trace then averages over all eigendirections rather than only the slowest.

D Cross-Domain Vision Experiment

We replicate the 2×2 ablation on CIFAR-100 [26] with a small Vision Transformer [13] (4 layers, 192 dim, 4 heads, patch size 4) and a RotNet auxiliary head [18] (4-way rotation prediction), using the same $\lambda = 0.05$ constant weight. Each configuration is trained for 200 epochs with AdamW [30] (lr$= 3{\times}10^{-4}$, cosine schedule) on a single GPU, with 5 seeds per cell.

Table 6: CIFAR-100 cross-domain test, 5 seeds per configuration. Means with population std. Final5 (mean of the last 5 epochs, the late-training metric matching Final50 in MARL) is the convention used in Table 5; Best is included for completeness.

Config	Encoder	Aux	Best	Final5	d vs Full (Final5)
Full	ViT	✓	43.94 ± 0.51	42.78 ± 0.27	–
No Aux	ViT	–	43.85 ± 0.51	43.18 ± 0.37	$+1.09$
No Attn	mean pool	✓	47.89 ± 0.12	47.08 ± 0.26	$+12.85$
Neither	mean pool	–	47.99 ± 0.13	46.83 ± 0.20	$+12.59$

Result. Under the late-training Final5 metric, Full vs No Aux gives Cohen's $d = +1.09$ (bootstrap 95% CI $[+0.04, +2.76]$, just resolved at n=5), corresponding to a ~ 0.4 percentage-point gap. This is two orders of magnitude smaller than the 8–34 point Best-to-Final50 drops observed in MARL, but it is not zero. We read this as a weak boundary residual rather than a clean null: the stationarity prediction holds in the strong sense that the mean-pooling vs attention encoder gap dominates the matrix (~ 4 points, an order of magnitude larger than the aux effect), but a careful late-training metric does not rule out a small remaining contribution from aux training stochasticity. Best accuracy gives a cleaner null ($d = -0.16$, CI $[-1.78, +1.19]$ crosses zero), reflecting that all four configurations reach comparable peak performance.

Proposition 1 predicts this: with stationary targets the target-generating policy has drift covariance $\Sigma_\pi = 0$, so $\Sigma_\varepsilon = J_\pi \Sigma_\pi J_\pi^\top = 0$ and the steady-state variance from Eq. (7) vanishes regardless of encoder type.

E Additional Tables and Per-Seed Data

Related observation in AERIAL. Independent of our design-pattern study, our actor-critic port of AERIAL [39] exhibits a consistent Best→Final50 drop on 4 of 5 seeds on Overcooked AA at 10M. The four converging seeds show a mean drop of 15.5 ± 3.3 points (per-seed Best minus Final50). Seed 1 fails to converge entirely, plateauing at ∼360. AERIAL has no auxiliary prediction head, so the aux-to-encoder pathway we isolate in VABL is not present; we do not claim our mechanism explains AERIAL's drop. Per-seed trajectories in Table 7.

Table 7: Per-seed AERIAL trajectories on Overcooked AA (10M steps).

Seed	Peak	Peak position	Final50	Drop
0	476.6	59%	460.1	16.5
1	361.5	100%	360.3	1.2‡
2	481.4	79%	468.5	12.9
3	478.0	42%	457.4	20.6
4	478.2	65%	466.2	12.0
4 healthy seeds	478.6 ± 1.7	–	463.0 ± 4.4	15.5 ± 3.3
All 5 seeds	455.1 ± 46.9	–	442.5 ± 41.3	12.6 ± 6.6

‡Seed 1 plateaued at ∼360 without a real peak. Our instability claims refer to the 4 healthy seeds.

Table 8: 6-method benchmark at 10M steps (5 seeds each). VABL uses the best ablation config (No Aux). All methods on Cramped Room saturate at the ceiling. Communication methods lead on AA in mean but with cross-seed std > 148, more than $40\times$ VABL's. These baselines are reported for context; they are not controls for the pathway-specific pathology since none instantiates the attention + teammate-prediction-aux design pattern.

		Asymm. Advantages		Cramped Room	
Method	**Comm?**	**Final50**	**std**	**Final50**	**std**
TarMAC	yes	**783.72**	156.10	414.73	2.78
CommNet	yes	773.24	148.17	416.30	0.43
MAPPO	no	548.23	153.09	**417.00**	**0.00**
VABL (No Aux)	no	473.45	**3.62**	416.18	0.71
AERIAL	no*	442.51	41.30	416.43	0.49
VABL (Full, pathology)	no	463.21	8.55	409.70	1.26

*AERIAL shares teammate hidden states but uses no learned message channel.

6-method benchmark (Overcooked AA and Cramped Room).

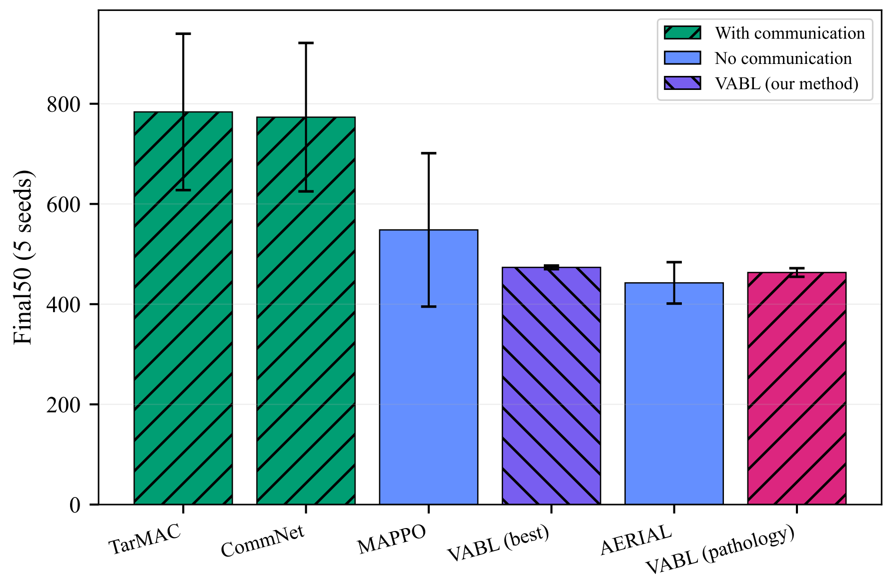


Figure 4: 6-method benchmark on Overcooked AA at 10M steps. Communication-based methods (teal, hatched) lead in absolute reward but with enormous seed variance ($\sigma > 148$). VABL (purple) has the tightest variance of any method ($\sigma < 4$). Data from Table 8 in the main body.

6-method benchmark figure (Overcooked AA).

Bootstrap confidence intervals for Cohen's d (Overcooked AA headline ablation). Cohen's d at n=5 is a point estimate with wide uncertainty. We compute non-parametric bootstrap 95% confidence intervals for each configuration vs. Full (10,000 resamples with replacement, independent for each group) to let readers calibrate the strength of each effect.

Config	Final50 mean $\pm$ std	d vs Full	95% bootstrap CI on d	Crosses zero?
Full (attn + aux)	463.20 ± 9.56	–	–	–
Neither (mean only)	469.72 ± 4.17	$+0.88$	$[-0.33, +2.73]$	yes
No Attention (mean + aux)	471.76 ± 2.08	$+1.24$	$[+0.31, +3.48]$	no
No Aux (attn only)	473.44 ± 4.02	$+1.40$	$[+0.49, +3.58]$	no
$+$ Anneal	473.66 ± 4.04	$+1.43$	$[+0.53, +3.55]$	no
$+$ Stop-gradient	471.60 ± 2.40	$+1.21$	$[+0.25, +3.45]$	no
$+$ Both	472.36 ± 2.47	$+1.31$	$[+0.42, +3.63]$	no

Table 9: Bootstrap 95% CI on Cohen's d vs. Full for Overcooked AA headline ablation. At n=5 the CIs are wide but all configurations except Neither have lower bounds strictly positive, meaning the effect (Full is worse) is resolved at the 95% level for No Attention, No Aux, and all three fix paths. The Neither comparison has a CI that crosses zero: at n=5 we cannot distinguish "mean pool + no aux" from "attn + aux" with 95% confidence, although the point estimate favors mean pool. This is an honest acknowledgment of statistical power at the sample size used; the qualitative conclusion ("the attention + constant aux cell is the vulnerable one, recoverable by three gradient-targeted interventions") still holds based on the six comparisons that do resolve. Std values in this table use the sample (Bessel-corrected, $n-1$) estimator; the main-body tables use the population estimator throughout.

	Aux ON (λ=0.05)	Aux OFF (λ=0)
Attention	463.21 ± 8.55	473.45 ± 3.62
Mean pool	471.76 ± 1.66	469.70 ± 3.33

Table 10: 2×2 interaction: Final50 mean $\pm$ std (5 seeds, 10M steps). The pathology occupies only the attention + aux-ON cell.

2×2 **interaction table (Overcooked AA).**

Config	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4
Full	469.1	456.5	450.6	465.6	474.2
Neither	473.5	474.8	467.4	465.3	467.6
No Attn	469.9	471.4	472.2	470.2	475.1
No Aux	475.9	475.1	477.8	469.8	468.6
B_anneal	478.9	469.4	472.4	470.8	476.8
B_stopgrad	472.6	471.7	472.1	467.6	474.0
B_anneal_sg	469.8	474.1	471.6	470.6	475.7

Table 11: Per-seed Final50 values for all 7 configurations on AA. Full VABL's range (23.6) is more than $2\times$ every alternative's (the largest alternative range is 9.5 for both Neither and B_anneal).

Per-seed Final50 (Overcooked AA, 7 configurations).

Config	Best	Final50	std	d vs Full
Full (attn + aux)	14.81	10.60	0.46	–
No Attention (mean + aux)	14.67	10.71	0.26	0.29
Neither (mean only)	14.33	10.20	0.44	-0.89
No Aux (attn only)	**15.69**	**11.15**	0.41	**1.26**
Fix paths (Full + gradient intervention)				
+ Anneal ($\lambda \to 0$)	14.83	10.63	**0.22**	$+0.10$
+ Stop-gradient	14.69	10.63	0.28	$+0.08$
+ Both	14.56	10.54	0.29	-0.16

Table 12: SMAX 3v3 ablation and fix paths (50K episodes, 5 seeds). The pathology reproduces: Full (10.60 ± 0.46) is beaten by No Aux (11.15 ± 0.41), $d = 1.26$. Fix paths reduce variance (std $0.46 \to 0.22$–0.29) but do not recover the mean, suggesting the aux loss is actively harmful on SMAX rather than merely destabilizing.

SMAX 3v3 ablation (4 configurations, 5 seeds).

Table 13: Per-seed Hanabi Final50 and peak score. Non-zero episode count (% of 25,024 episodes scoring non-zero reward) indicates training stability: Aux-ON cells (Full, No Attn) spend most episodes at 0, while Aux-OFF cells (No Aux, Neither) sustain positive scoring. Seed 3 of Full scores only 1 non-zero episode in 25K; this is a complete training collapse, not a gradual drift.

Config	Seed	Final50	Best	Non-zero eps %
Full	0	0.60	7.0	3.3%
Full	1	0.00	9.0	3.2%
Full	2	0.00	7.0	23.2%
Full	3	0.00	5.0	$< 0.01\%$
Full	4	0.08	6.0	3.4%
No Aux	0	3.64	7.0	44.6%
No Aux	1	0.80	7.0	68.7%
No Aux	2	3.34	7.0	65.8%
No Aux	3	4.06	9.0	77.9%
No Aux	4	2.20	10.0	43.6%
No Attn	0	0.48	7.0	10.5%
No Attn	1	0.00	7.0	2.6%
No Attn	2	0.66	7.0	34.7%
No Attn	3	0.16	6.0	0.6%
No Attn	4	0.00	3.0	$< 0.01\%$
Neither	0	1.94	9.0	31.5%
Neither	1	3.02	7.0	63.0%
Neither	2	0.44	8.0	19.2%
Neither	3	3.80	8.0	54.5%
Neither	4	0.02	7.0	0.1%

Hanabi 2-player per-seed data (5 seeds per config, 25K episodes, horizon 80).

Table 14: Per-seed gradient decomposition statistics on Overcooked AA. Cosine std is over the late 50% of training iterations (8 log points per seed, logged every 25 iterations). $\|g_\pi\|$ and $\|g_{\mathrm{aux}}\|$ are means over logged iterations. Under No Aux the aux gradient is identically zero; under Stop-grad and late-stage Anneal the aux gradient has zero projection onto policy-gradient-receiving parameters (disjoint param supports), giving cosine exactly zero by construction. Only Full exhibits non-zero cosine temporal variance, consistent with directional aux-policy conflict.

Config	Seed	cosine std (late 50%)	cosine mean (late)	$\|g_\pi\|$	$\|g_{\mathrm{aux}}\|$
Full	0	0.196	$+0.086$	0.117	0.016
Full	1	0.164	$+0.022$	0.140	0.016
Full	2	0.226	$+0.029$	0.144	0.016
Full	3	0.175	-0.116	0.129	0.016
Full	4	0.135	-0.000	0.114	0.014
No Aux	all	0.000	0.000	0.118–0.137	0.000
Stop-grad	all	$0.000^\dagger$	$0.000^\dagger$	0.110–0.150	0.010–0.013
Anneal	all	$0.000^\dagger$	$0.000^\dagger$	0.115–0.143	0.004–0.006

†Exact zero by construction: under stop-gradient the aux-to-encoder pathway is cut and g_{aux} has support only on aux-MLP parameters (disjoint from policy-gradient support); under anneal late training has $\lambda \to 0$, so $g_{\mathrm{aux}} \to 0$. The Full signature is the only dynamically-generated cosine variance.

Gradient decomposition per-seed detail (Full VABL + fix paths, 5 seeds).

Table 15: Per-seed Final50 across four MARL environments and both primary configs (Full vs No Aux). Seed columns are the 5 independent runs per cell; d column is Cohen's d (No Aux vs Full, pooled sample std).

Env	Config	s0	s1	s2	s3	s4	mean	d
AA	Full	469.12	456.50	450.64	465.56	474.24	463.21	—
AA	No Aux	475.90	475.10	477.80	469.80	468.60	473.45	+1.40
Cramped	Full	410.25	407.85	411.34	409.14	409.92	409.70	—
Cramped	No Aux	416.31	416.95	415.37	417.04	415.24	416.18	+5.07
SMAX	Full	10.82	10.71	11.16	10.13	10.20	10.60	—
SMAX	No Aux	11.47	10.94	11.55	10.78	11.00	11.15	+1.26
Hanabi	Full	0.60	0.00	0.00	0.00	0.08	0.14	—
Hanabi	No Aux	3.64	0.80	3.34	4.06	2.20	2.81	+2.81

Cross-environment per-seed summary.

F Proxy–Variance Plot

This appendix supports the "From cosine to proxy" paragraph in §4. Recall that linearizing $\cos(g_\pi, g_{\text{aux}})$ at fixed norms gives $\text{Var}_t[\cos] \approx \hat{g}_\pi^\top \Sigma_\varepsilon \hat{g}_\pi / \|\bar{g}_{\text{aux}}\|^2$, so the cosine temporal standard deviation is a finite-sample proxy for normalized $\text{tr}(J_\pi \Sigma_\pi J_\pi^\top)^{1/2}$. If the theory is correct, this proxy should predict the Final50 cross-seed standard deviation ordering.

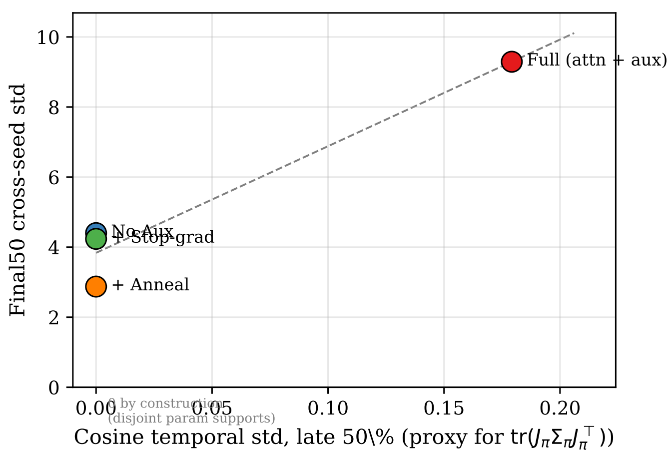


Figure 5: Cosine temporal std (proxy for $\text{tr}(J_\pi \Sigma_\pi J_\pi^\top)$ normalized by gradient magnitudes) vs Final50 cross-seed std across four logged conditions on Overcooked AA. Three zero-proxy conditions reach zero by pathway construction; Full is the only condition with non-zero proxy and has elevated variance. Linear $R^2 = 0.94$ across conditions; the ordering is the main prediction, not the slope.

How to read the plot. Each point is one of the four logged conditions from ExpB (Full, No Aux, Stop-grad, Anneal), aggregated over 5 seeds. The x-axis is the cosine temporal standard deviation measured in the late 50% training window; the y-axis is the Final50 cross-seed standard deviation. Three of the four conditions have proxy *exactly zero* by construction: No Aux has no aux gradient so $g_{\text{aux}} = 0$; Stop-grad severs the encoder-subspace projection of J_π; Anneal drives $\lambda \to 0$ in the late window. Only Full retains a non-zero proxy, and it is the only condition with elevated Final50 variance.

What the fit verifies. The $R^2 = 0.94$ linear fit confirms the ordering prediction: the condition with non-zero proxy ($J_\pi \cdot \Sigma_\pi$ both non-zero) has the highest cross-seed variance, and the three conditions with zero proxy cluster together at low variance. Since three of four x-values are identically zero, the fit is dominated by the categorical separation rather than a continuous slope. Within Full's 5 seeds, per-seed cosine std correlates with per-seed peak-to-Final50 drop at $r = +0.40$, in the predicted direction but weak at $n = 5$.

Scope. The proxy is a first-order approximation and shares the scope limits of Proposition 1 (Appendix A.5). In particular, the normalization by $\|\bar{g}_{\text{aux}}\|^2$ means the proxy measures the *direction* of the aux gradient relative to the policy gradient, not the aux gradient's absolute energy; a condition with a large-magnitude but direction-stable aux gradient would read as low proxy without necessarily being free of interference.

G Synthetic Verification of Proposition 1

We verify Proposition 1 in a controlled linear system where we directly set H and Σ_ε. Figure 6 shows the empirical steady-state variance (blue) matches the theoretical prediction (red dashed) across sweeps of $\operatorname{tr}(\Sigma_\varepsilon)$, $\lambda_{\min}(H)$, and α. Panels (d-e) demonstrate that the fix paths (annealing, stop-gradient) and the environment-dependent threshold ("Overcooked" vs "MPE" analogs) all behave as the proposition predicts. The variance scales linearly with the critical ratio $\operatorname{tr}(\Sigma_\varepsilon)/\lambda_{\min}(H)$ (panel f).

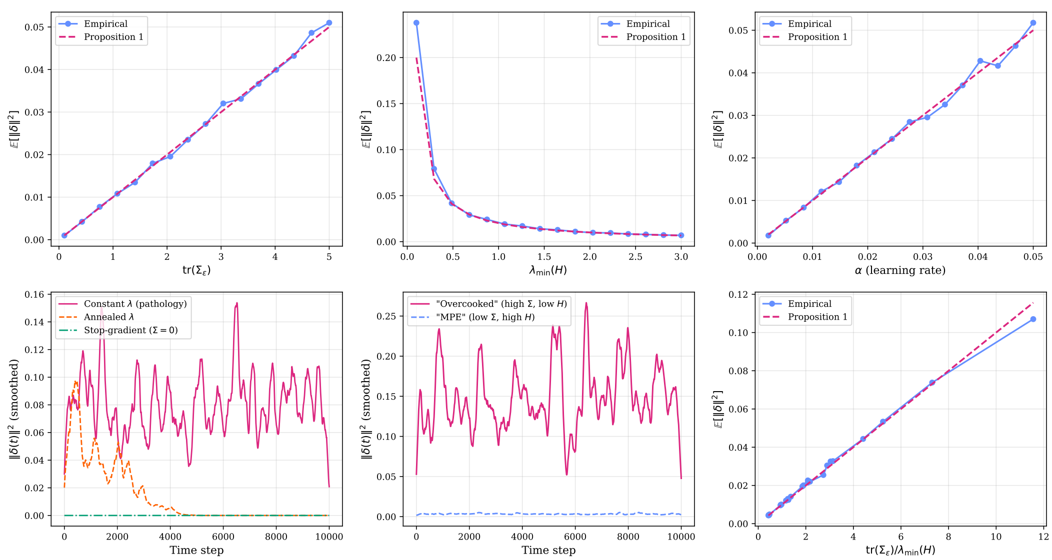


Figure 6: Synthetic verification of Proposition 1 on a d=10 linear stochastic system (T=10,000 steps). Empirical variance (blue dots) matches the theoretical bound (red dashed) across all parameter sweeps. Panel (d): fix paths recover stability. Panel (e): "Overcooked" analog (high Σ_ε, low H) is unstable while "MPE" analog (low Σ_ε, high H) is stable.

H λ Sensitivity (Overcooked AA)

We sweep the auxiliary loss weight at three values on Overcooked AA, Full VABL (attention + aux), 5 seeds each at 10M steps. Two of the values (λ=0.01 and λ=0.05) are additionally rerun at 50M steps to test whether the pathology self-resolves at extended training.

Table 16: λ sensitivity on Overcooked AA (Full VABL, 5 seeds). Final50 mean $\pm$ std. λ=0.01 is the stable sweet spot; λ=0.05 exhibits the directional-interference pathology; λ=0.001 has a comparable mean but elevated cross-seed variance at this sample size.

λ	Budget	Final50	std	Best	Best std
0.001	10M	468.00	6.37	484.40	1.20
0.01	10M	472.23	1.67	484.40	1.20
0.05	10M	463.21	8.55	483.80	1.47
0.01	50M	468.47	8.73	485.00	0.00
0.05	50M	470.95	7.46	485.00	0.00

Table 17: Per-seed Final50 for the λ sweep at 10M on Overcooked AA. Best is reached consistently at $\sim$484 across all λ; only Final50 differentiates, confirming that λ affects late-training stability rather than peak performance.

λ	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4
0.001	475.3	461.0	471.3	459.7	472.7
0.01	475.1	472.0	470.0	472.7	471.4
0.05	469.1	456.5	450.6	465.6	474.2

At 50M steps, both λ=0.01 and λ=0.05 retain cross-seed std above 7, confirming that the pathology is not a transient artifact of the 10M budget and does not self-resolve with $5\times$ more training. Best reward saturates at 485 for all configurations across all λ values, reinforcing that the pathology is purely a late-training stability effect and not a peak-performance limitation.

I Architecture Sweep

We test the pathology across 6 architecture variants by swapping recurrence (GRU [9], LSTM [21], or feedforward) and attention (cross-attention, self-attention, additive [1], or mean pooling) independently while keeping all other hyperparameters fixed. Each configuration is run with and without the auxiliary loss (5 seeds each, 60 runs total). A MAAC-style variant places the auxiliary head on the centralized critic instead of the actor.

Table 18: Architecture sweep on Overcooked AA (25K episodes, 5 seeds each). Cohen's d: positive means aux hurts. The pathology generalizes across recurrence types and attention mechanisms but is absent with mean pooling and reverses when aux targets the critic.

Recurrence	**Attention**	**Aux loc.**	**+aux Final50**	**$-$aux Final50**	d	**Pathology?**
LSTM	cross-attn	actor	436.5 ± 22.1	454.0 ± 21.2	$+0.80$	**yes**
GRU	additive	actor	440.4 ± 21.8	524.8 ± 116.5	$+1.01$	**yes**
GRU	self-attn	actor	527.6 ± 145.2	544.4 ± 156.1	$+0.11$	no‡
GRU	mean pool	actor	431.7 ± 25.3	422.8 ± 33.2	-0.30	no
FF (none)	cross-attn	actor	588.1 ± 195.3	609.6 ± 208.7	$+0.11$	no‡
GRU	cross-attn	**critic**	546.6 ± 156.6	458.1 ± 21.5	-0.79	**reversed**

‡Both conditions exhibit high intrinsic cross-seed variance ($\sigma > 145$) in both the +aux and $-$aux cells, making the aux effect undetectable.

The pathology appears whenever *any* recurrence (GRU or LSTM) is paired with *any* learned attention mechanism (cross-attention or additive) and a constant auxiliary loss on the actor. Mean pooling, which has no learned attention parameters, is immune ($d = -0.30$). The MAAC-style critic-auxiliary configuration reverses the sign of the mean effect ($d = -0.79$, with No Aux below the +aux cell), but the cross-seed std also rises sharply (from 21.5 to 156.6), so this is a qualitatively different regime — higher mean, much higher variance — rather than improved stability under our variance-based definition. We mark this row **reversed** (sign of the mean effect, not stability) and place it outside the scope of Proposition 1.

J Environments, Baselines, and Metrics

This appendix expands §5's parenthetical summary of the experimental infrastructure.

J.1 Environments

We evaluate four MARL environments via JaxMARL [41] and one supervised control.

Overcooked [6]: *Asymmetric Advantages* and *Cramped Room*, 2 agents, horizon 400, 25,024 episodes ($\approx$10M steps), 64 vectorized envs.

Hanabi 2-player [2]: turn-based partial-information card game, horizon 80, 25K episodes. Absolute scores at this horizon are low; the relative ordering, not the magnitudes, is the claim.

MPE simple_spread [31]: 3 symmetric agents, horizon 25, 100K episodes.

SMAX 3v3 [41]: 3 allies versus 3 heuristic enemies, horizon 100, 50K episodes, covering both SMAC [42] and SMAC v2 [15] scenarios (tactical combat).

CIFAR-100 [26] (supervised control): small Vision Transformer [13] (4 layers, 192 dim, 4 heads, patch size 4) with a RotNet auxiliary head [18] (4-way rotation prediction), $\lambda = 0.05$ constant, 200 epochs, AdamW with cosine schedule.

J.2 Baselines

We compare against four methods spanning three paradigms.

MAPPO [58]: decentralized policy gradient with centralized critic; the standard cooperative-MARL baseline.

TarMAC [12]: targeted attention-based inter-agent communication.

CommNet [46]: broadcast communication via hidden-state averaging.

AERIAL [39]: attention over teammate hidden states; our port uses PPO for comparability with the rest of the matrix, whereas the original is QMIX-based.

All baselines are reimplemented in JAX [4] using JaxMARL [41] with matched hyperparameters, env wrapper, and evaluation protocol, so cross-method differences reflect algorithmic choices rather than infrastructure drift. None of these baselines instantiates the attention + teammate-prediction-aux design pattern; they are reported for context rather than as controls for the core mechanism. Direct re-implementation of Dynamic Belief [60] or BEPAL [22], which do instantiate the pattern, is left to future work; we rely on Zhai et al.'s independently-reported Static-Belief failure mode as external corroboration (§2).

J.3 Metrics

- **Best**: maximum episode reward (or maximum test accuracy in the supervised control) achieved during training.
- **Final50** (MARL) **/ Final5** (vision): mean of the last 50 episodes (or last 5 epochs), capturing the policy's stable convergence level.
- **Collapse %**: $(\mathrm{Best} - \mathrm{Final})/\mathrm{Best} \times 100$, used only when discussing the broader literature's coordination-collapse framing.
- **Cross-seed std**: population standard deviation across the 5 random seeds per cell; the headline stability metric for our pathology.
- **Cohen's** d [11] with bootstrap 95% confidence intervals ($n_{\mathrm{boot}} = 10{,}000$) for effect-size comparisons. A CI that crosses zero signals a non-resolvable effect at $n = 5$.

All MARL numbers report mean $\pm$ std over 5 random seeds unless otherwise noted.

K Hyperparameters

Parameter	Value
Architecture	
Observation embedding dim d_e	64
GRU hidden / belief dim d_h	128
Attention heads (MHA)	4
Aux predictor MLP	2 layers, hidden 64, ReLU
Centralized critic MLP	2 layers, hidden 128, ReLU
Action embedding dim	64
PPO [44]	
Clip ϵ	0.2
Epochs per update	10
Mini-batches per epoch	4
GAE λ [43]	0.95
Discount γ	0.99
Value-loss coefficient c_v	0.5
Entropy coefficient c_e	0.01
Optimization	
Optimizer	Adam [25], $\epsilon = 10^{-5}$
Learning rate (actor, critic)	5×10^{-4}
Gradient clip (global norm)	10.0
Auxiliary loss (when enabled)	
Aux lambda λ	0.05 (constant unless annealed)
Anneal schedule (when used)	linear $\lambda \to 0$ over first 50% of training
Environment	
Parallel envs (JAX vmap)	64 (Overcooked, SMAX) / 32 (Hanabi)
Horizon (Overcooked)	400
Horizon (MPE)	25
Horizon (Hanabi)	80
Horizon (SMAX)	100

Table 19: Shared hyperparameters across all VABL configurations and baselines on the MARL benchmarks.

CIFAR-100 cross-domain control. Architecture: small Vision Transformer with 4 transformer blocks, embedding dim 192, 4 attention heads, MLP ratio 4.0, patch size 4 (64 patches per 32×32 image plus a CLS token). RotNet auxiliary head: 2-layer MLP with hidden 192 and GELU activations, predicting one of 4 rotation angles ($0°, 90°, 180°, 270°$). Training: 200 epochs, batch size 256, AdamW with learning rate 3×10^{-4}, weight decay 0.05, cosine schedule with 5 warm-up epochs, gradient clip 1.0. Standard CIFAR-100 augmentation (random crop with padding 4, random horizontal flip, mean/std normalization). Auxiliary lambda matches the MARL setting ($\lambda = 0.05$). The mean-pooling variant replaces every $\mathrm{MultiheadAttention}$ block with a per-token MLP (hidden dim equal to embedding dim) so that parameter counts are approximately matched between the attention and mean-pool conditions.

L Figure and Reader Accessibility

Reader accessibility. The paper is structured for a reviewer scan: the title opens with a curiosity prompt before stating the mechanism; the abstract puts problem, mechanism, and what-the-failure-is-not in three citation-free sentences; an italicized one-line principle closes the first intro paragraph; Figure 1 gives a four-box conceptual chain (target-policy updates $\to \Sigma_\pi \to \Sigma_\varepsilon = J_\pi \Sigma_\pi J_\pi^\top \to$ instability) before any math; contributions are a scannable bulleted list with noun-phrase labels; and the Theory section opens with a plain-language intuition sentence before the four-step outline. Numeric thresholds and bootstrap CIs are deferred from the introduction to Methodology and Results, where they carry their full meaning.

Figure accessibility. All figures in this paper are designed for accessibility across viewing conditions. One author has color vision deficiency, which motivated a systematic approach to figure design:

- **Color palette.** We use the IBM colorblind-safe palette (magenta, blue, purple, orange, gold, teal), which maintains perceptual distinctness under deuteranopia, protanopia, and tritanopia. The pathological configuration (Full VABL) uses magenta throughout all figures for consistent identification.
- **Redundant encoding.** Every data series is distinguished by at least two independent visual channels. Dot plots use distinct marker shapes (cross, square, diamond, circle). Bar charts use hatching patterns (diagonal, dots, backslash, crosshatch) in addition to color. Line plots use distinct line styles (solid, dashed, dash-dot, dotted) with large markers.
- **Grayscale readability.** All figures remain interpretable when printed in black-and-white. We verified this by converting each figure to grayscale and confirming that all data series remain distinguishable through shape, pattern, and line style alone.
- **Font consistency.** Figure text (axis labels, legends, annotations) uses the same serif font family as the paper body to maintain visual coherence. We avoid matplotlib's default sans-serif rendering.
- **No duplicate titles.** Figures use LaTeX captions only, with no redundant matplotlib titles baked into the image. This avoids visual clutter and ensures consistent typography.

We encourage the MARL community to adopt similar accessibility practices. Approximately 8% of males and 0.5% of females have some form of color vision deficiency, and many readers access papers in printed grayscale. Relying solely on color to distinguish data series excludes these readers.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and §1 claim a learning-dynamics principle (structured non-stationary aux targets ⇒ directional gradient noise) and enumerate contributions. §4 formalizes the principle and §6 tests it.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: §7 has a dedicated Limitations section covering first-order model assumptions, the critic-side reversal, the SMAX residual mean gap, empirical scale ($n{=}5$ per cell), and future-work items.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions.
   - The authors should reflect on the scope of the claims made.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Proposition 1 states all assumptions inline (local quadratic loss, i.i.d. noise, spectral-radius contraction). The full derivation is Appendix C. Step-by-step context, colored-noise correction, linearization scope, and the λ-prediction scope are in Appendix A. All equations are numbered and cross-referenced.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.

- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

   Answer: [Yes]

   Justification: §5 specifies the ablation matrix, fix-path interventions, and target-source distinguishing test. Appendix J gives full environment specifications, baselines, and metric definitions. Appendix K lists all shared hyperparameters. All experiments use 5 seeds with bootstrap CIs. The code and configs are included in the supplementary archive.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The supplementary archive contains the full code, the per-seed result JSONs for every experiment, a top-level reproducibility driver, and per-experiment launch scripts. Data dependencies (JaxMARL environments, CIFAR-100 via torchvision) are open-source and loaded via standard pip installs.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines.
   - The instructions should contain the exact command and environment needed to run to reproduce the results.
   - At submission time, to preserve anonymity, the authors should release anonymized versions.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: §5 covers the ablation matrix; Appendix J provides full environment and baseline specs; Appendix K lists PPO clip, GAE λ, optimizer, discount, network dimensions, horizons per environment, and all VABL-specific knobs. The λ sweep (Appendix H) and architecture sweep (Appendix I) explicitly document hyperparameter selection.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: All MARL and CIFAR-100 cells report mean $\pm$ population std over 5 random seeds. Effect sizes use Cohen's d; Appendix J.3 for the full metric definition.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests.
- The factors of variability that the error bars are capturing should be clearly stated.
- The method for calculating the error bars should be explained.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: All experiments were run on a single-GPU workstation with an NVIDIA RTX 5090. A typical VABL 10M-step Overcooked run takes $\sim$2 hours; the CIFAR-100 4×5-seed matrix takes $\sim$7.3 hours total; the full 290+-run matrix is $\sim$400 GPU-hours end-to-end. No cluster or distributed setup is required, although it might help running parallely; the supplementary archive contains per-experiment launch scripts and a top-level reproducibility driver.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The research studies gradient dynamics in cooperative multi-agent RL using standard synthetic environments (Overcooked, Hanabi, MPE, SMAX) and a public image dataset (CIFAR-100). No human subjects, no privacy-sensitive data, no dual-use model release. Anonymity is preserved in the submission and linked repository.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: This is a theoretical and empirical paper on gradient dynamics of auxiliary-loss training in cooperative reinforcement learning. It does not propose a deployable model, training recipe for a production system, or dataset with differential access risks. We see no direct societal-impact pathway distinct from the general impact of MARL research.

    Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- The conference expects that many papers will be foundational research and not tied to particular applications.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The paper does not release pretrained models, generative models, or scraped datasets. The only released artifacts are training code and result JSONs, both of which pose no identifiable misuse risk.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: JaxMARL [41] (Apache 2.0), Overcooked-AI [6] (MIT), SMAC/SMACv2 [42, 15] (MIT), Hanabi [2] (Apache 2.0), CIFAR-100 [26] (research use). All are cited in Related Work and Appendix J.1; our use complies with each license.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The name of the license should be included for each asset.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The VABL implementation, all experiment scripts, and the complete results matrix (160+ per-seed JSON files) are included in the supplementary archive. The archive contains a README, a top-level reproducibility driver, per-experiment launch scripts, analysis scripts, and plotting code. Intended use, scope, and known limitations are documented alongside the code.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates.
    - At submission time, remember to anonymize your assets.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: No crowdsourcing or human-subject research. All data come from simulated environments and a public image dataset.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: No human subjects, so no IRB review is applicable.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

    Answer: [N/A]

    Justification: LLMs were used only for writing, editing, and formatting assistance. They are not a component of the core methodology, theoretical derivation, or experimental setup.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.